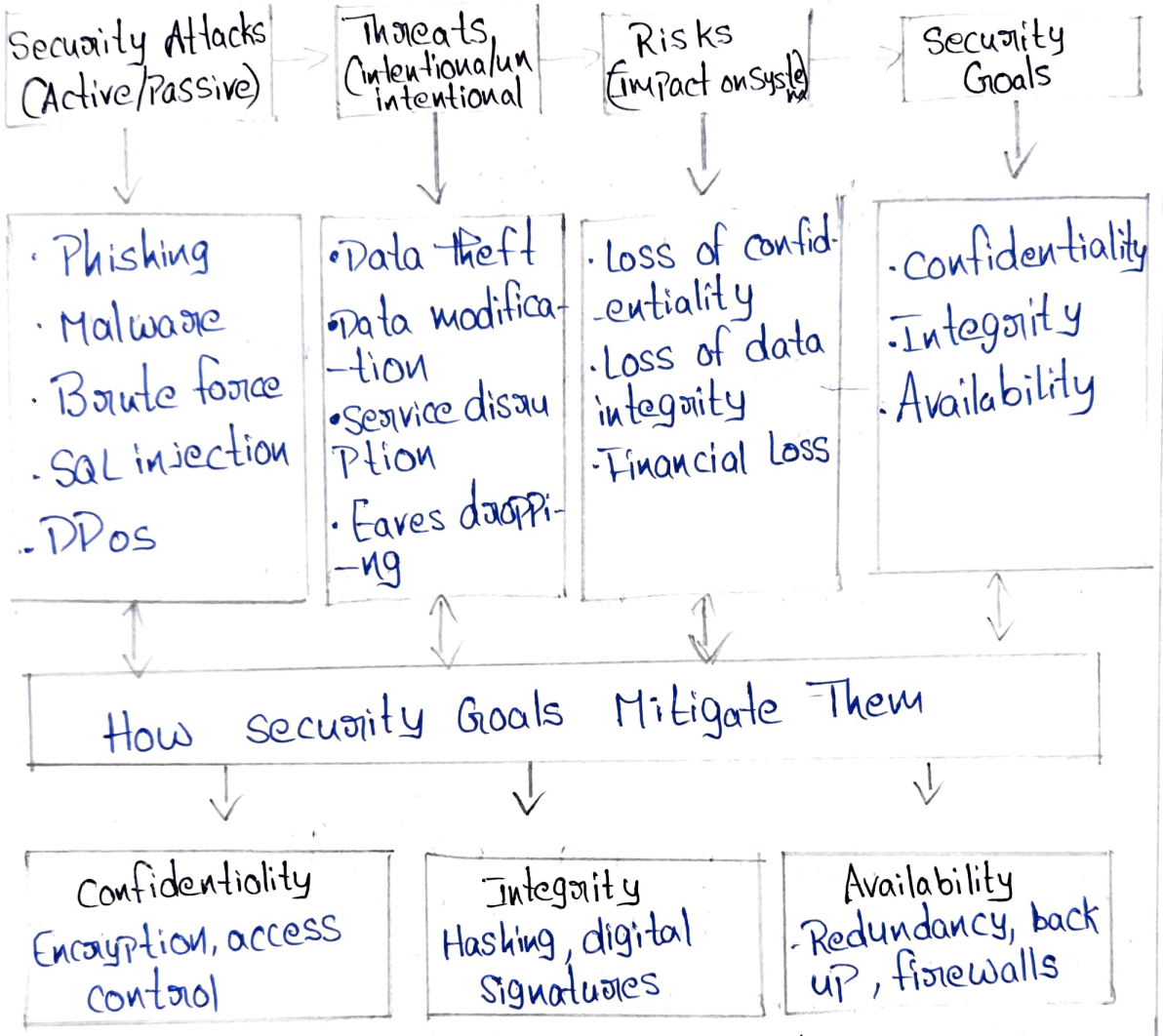
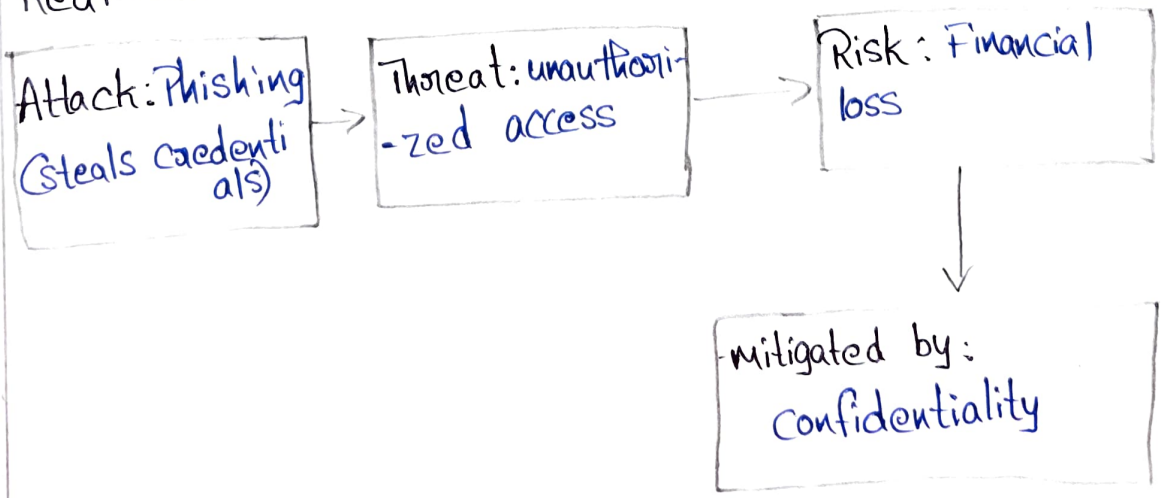
Assessment Tool - 2

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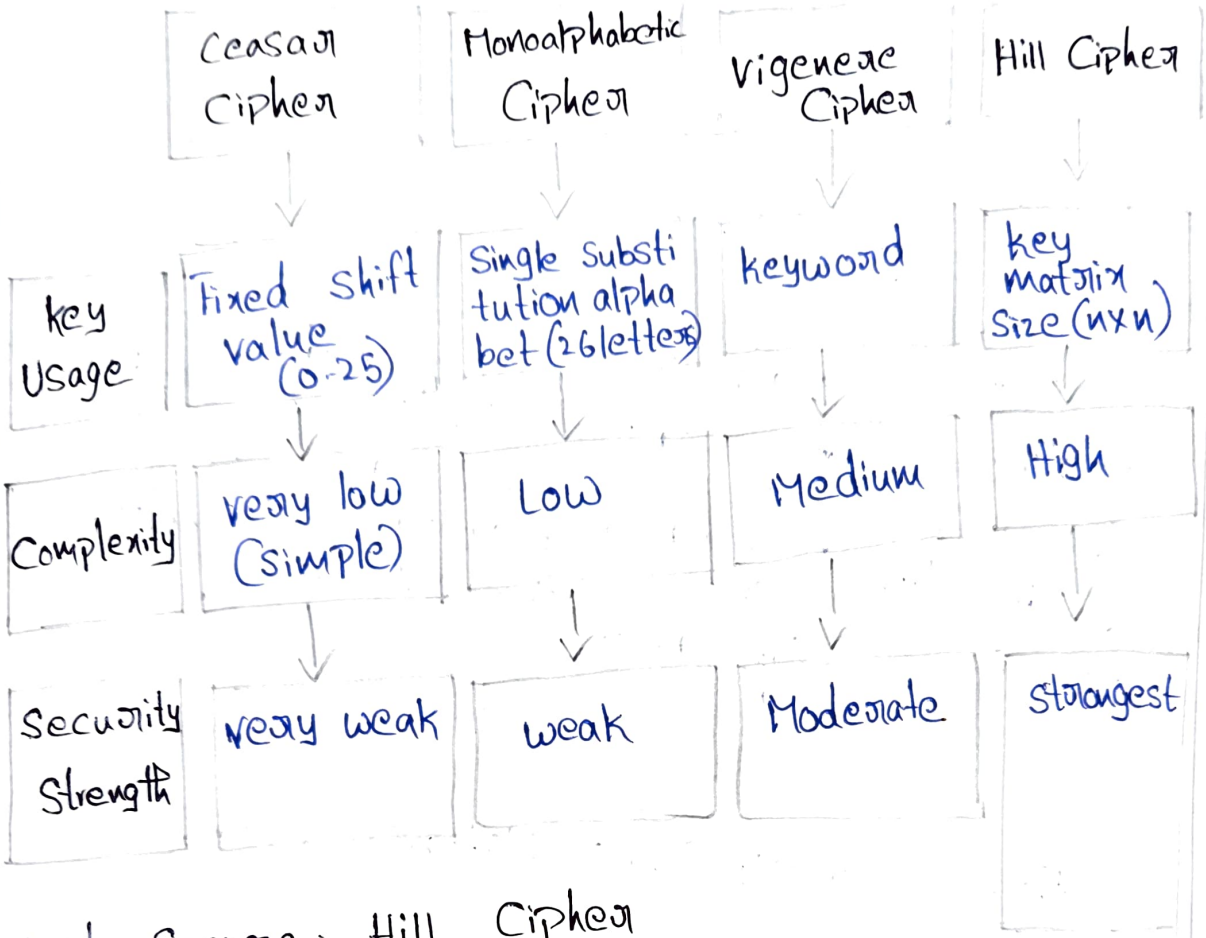
1. Security Attacks - Threats - Risks - Security Goals



Real-world example for each link



9. Substitution Techniques Comparision



Most Secure: Hill Cipher

Uses a matrix key and block processing. which makes it much harder to break with simple frequency analysis or brute force compared to the other substitution techniques

3 Classical Encryption : Substitution Vs Transposition

Classical Encryption Schemas

Substitution Techniques

Transposition Techniques

Differences and Similarities

Aspect	Substitution	Transposition	Similarity
Data Structure	Changes the actual characters (A→D)	Keeps characters same, order	Both are classical
Security level	Lower	Slightly higher	Both are weaker than modern encryption
Key Usage	Alphabet/ keyword	Permutation key	Both require key
Complexity	Low to High	Medium	Both are simple compared to modern ciphers.

Substitution techniques are generally more vulnerable to attacks because frequency analysis can often reveal Patterns. Transposition is harder to break as it preserves.

Steganography

Data Hiding
Hides information within another medium

Watermarking
Embeds information to prove ownership to authenticity

Survivability
Ability of hidden data to remain even if the medium is altered

How they Contribute to Secure Communication

Data Hiding
→ keeps Communication secret from another Parties

Watermarking
→ Adds authenticity and protects against tampering

Survivability
Ensures the hidden data is robust data is robust and can be recovered even after changes

Real-World Applications

Data Hiding
Secret messages in images

Watermarking
copyright Protection in digital media

Survivability
Robust watermarks in videos that Survive compression and noise